

Seminar on Artificial Intelligence & Quantum Computing



Instituto Politécnico Nacional
"La Técnica al Servicio de la Patria"



OCTOBER 16 & 17 2025

Free Admission

Program

Thursday, October 16, 2025

9:00	REGISTRATION DAY 1
9:30	WELCOME – Oscar Montiel Ross
9:40	Lightweight Graph-Based Methods for Sea Turtle Identification Jesus Fernando Franco-Lopez, Alejandro Alvaro Ramirez-Acosta, Mireya S Garcia-Vazquez Centro de Investigación y Desarrollo de Tecnología Digital (CITEDI) IPN Corr. author: msarai@ipn.mx
10:00	Speaker Identification and Gender Recognition: A Benchmarking Study of ML and DL Models Ricardo Rioda Santiago, <u>Daniel Santiago Aguila-Torres</u> , Raul Rodriguez, Miguel Lopez-Montiel ITJuana Labs Corr. author: miguel.lopez@itj.com
10:20	Towards Practical Implementation of Quantum Key Distribution for Secure Communication in the Post-Quantum Era Seyi Tope Ogunji, Moises Sánchez-Adame, Oscar Montiel Ross Centro de Investigación y Desarrollo de Tecnología Digital (CITEDI) IPN Corr. author: oross@ipn.mx
10:40	Cucumber disease inspection for automated quality control in an agricultural produce facility based on deep learning Kenia Picos, Ulises Orozco-Rosas Centro de Enseñanza Técnica y Superior (CETYS) Campus Tijuana Corr. author: kenia.picos@cetys.mx
11:00	 Keynote – Fractal Search for Topological Optimization of Neural Network Architectures Patricia Melin, Oscar Castillo, Marylu L. Lagunes, Ivette Miramontes Instituto Tecnológico de Tijuana (ITT) Corr. author: pmelin@tectijuana.mx
11:40	COFFEE BREAK
12:00	 Keynote – Optimization of Type-2 Fuzzy Systems and Future Trends for Type-3: Theory and Application Oscar Castillo Instituto Tecnológico de Tijuana (ITT) Corr. author: ocastillo@tectijuana.mx

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| 12:40 | Variational Quantum Eigensolver for Clinical Biomarker Classification: A Hybrid Quantum-Classical Approach to HIV/TB Co-Infection and Drug Resistance Prediction
Juan Pablo Acuña González, Moisés Sánchez-Adame, Oscar Montiel
Centro de Investigación y Desarrollo de Tecnología Digital (CITEDIPN)
Corr. author: oross@citedi.mx |
| 13:00 | Artificial Intelligence-Based Image Processing for Osteoporosis Detection in Panoramic Radiograph
Montserrat Marrón-Castillo, Wendy Flores-Fuentes, Juan C. García-Gallegos, Julio C. Rodríguez-Quirón.
Universidad Autónoma de Baja California (UABC) Campus Mexicali
Corr. author: anabel.martinez@upp.edu.mx |
| 13:20 | Integrating Quantum Intermediate Representation into a C-Based Quantum-Classical Framework for Efficient Quantum Computing
Paul O. Onoja, Seyi Tope Ogunji, Moises Sanchez-Adame, Oscar Montiel Ross, Ushnik Nath
Centro de Investigación y Desarrollo de Tecnología Digital (CITEDIPN)
Corr. author: msanchez@citedi.mx |
| 13:40 | The need of quantum computing for Fuzzy Systems in high dimensionality problems
Prometeo Cortés-Antonio, Fevrier Valdez, Patricia Melin, Oscar Castillo
Instituto Tecnológico de Tijuana (ITT)
Corr. author: cortes@tectijuana.edu.mx |
| 14:00 | Fuzzy Dragonfly Algorithm for Solving Global Optimization Problems
Fevrier Valdez and <u>Héctor M. Guajardo</u>
Instituto Tecnológico de Tijuana (ITT)
Corr. author: fevrier@tectijuana.mx |
| 14:20 | Solving the 5-queens problem using the Grover's algorithm
Daniel Alejandro López Montiel
Centro de Investigación y Desarrollo de Tecnología Digital (CITEDIPN)
Corr. author: dlopez@citedi.mx |
| 14:40 | Deep Learning-Based Image Colorization Using Generative Models
Vannesa Argueta, Ulises Orozco-Rosas, Kenia Picos
Centro de Enseñanza Técnica y Superior (CETYS) Campus Tijuana
Corr. author: ulises.orozco@cetys.mx |
| 15:00 | CLOSING DAY 1 |

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9:00	REGISTRATION DAY 2
9:30	WELCOME – Oscar Montiel Ross
9:40	GPU-Accelerated PSO-APF Learning Algorithm for Mobile Robot Path Planning Ulises Orozco-Rosas, Kenia Picos, Oscar Montiel Centro de Enseñanza Técnica y Superior (CETYS) Campus Tijuana Corr. author: ulises.orozco@cetys.mx
10:00	Analysis of quantum approximation algorithms for combinatorial optimization problems Jorge Saavedra Benavides, Arturo Alejandro Arvizu Velazquez, Jesús Yaljá Montiel Pérez, Juan Humberto Sossa Azuela Centro de Investigación en Computación (CIC) IPN Corr. author: jsaavedrab2024@cic.ipn.mx
10:20	Reinforcement Learning Tuner for Enhancing Decision-Making in Exploring a solution space of the Electric Fleet Size and Mix Vehicle Routing Problem with Time Windows Alma Danisa Romero Ocaño, Víctor Manuel Valenzuela Alcaraz, Anabel Martínez Vargas Universidad Politécnica de Pachuca (UPP) Corr. author: ma.cosio.leon@upp.edu.mx
10:40	Leveraging Generative Adversarial Networks for Synthetic Data Generation: Applications in Tabular Data Ricardo Rioda Santiago, <u>Yahir Soto</u> , Emmanuel Cardenas, Miguel Lopez-Montiel ITJuana Labs Corr. author: miguel.lopez@itj.com
11:00	Keynote – Artificial Neural Networks: Basics and why they are so powerful Humberto Sossa Centro de Investigación en Computación (CIC) IPN Corr. author: humbertosossa@gmail.com
11:40	Quantum Deep Q-Networks with Variational Circuits: A Hybrid Quantum-Classical Approach to Reinforcement Learning Jonathan Oswaldo Trejo Reyes, Alejandro Monroy Azpeitia, Rolando Menchaca Mendez, Alberto Maldonado Romo, Jesus Yalja Montiel Perez Centro de Investigación en Computación (CIC) IPN Corr. author: trejor2023@cic.ipn.mx
12:00	COFFEE BREAK

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| 12:20 | Resource-Efficient CNNs for TIG Welding Quality Monitoring Using Data Augmentation, Undersampling, and Class-Balanced Loss Techniques
Ricardo Rioda Santiago, <u>Antonio Contreras Ortiz</u> , Nicolas Septien, Miguel Lopez-Montiel
ITJuana Labs
<i>Corr. author: miguel.lopez@itj.com</i> |
| 12:40 | Automatic Reviewer of Handwritten Flowchart Structures Using Convolutional Neural Networks
Josue Dominguez Guerrero, Alexa Daniela Aguirre Alvarez, Rene Noriega Armendariz, Everardo Santiago Ramirez
Universidad Autonoma de Ciudad Juarez (UACJ)
<i>Corr. author: josue.dominguez@uacj.mx</i> |
| 13:00 | A novel approach toward Adiabatic Quantum Computing to tackle 3-SAT problem
Sandra S. Rosales-Alvarado, Oscar Montiel, Ulises Orozco-Rosas
Centro de Investigación y Desarrollo de Tecnología Digital (CITEDI) IPN
<i>Corr. author: oross@ipn.mx</i> |
| 13:20 | Automated decision-support system for dynamic data acquisition and analysis in cryptocurrency markets
Selene Solis Arellano, Ulises Orozco-Rosas
Centro de Enseñanza Técnica y Superior (CETYS) Campus Tijuana
<i>Corr. author: ulises.orozco@cetys.mx</i> |
| 13:40 | Integration of Quantum Techniques and Artificial Intelligence in the Adjustment of Optical Aberrations using Shack-Hartmann Sensors
Mario Alberto Jara Reyna, Carlos Alberto Guerrero Peña, Juan José Tapia Armenta, Joel Herrera Vázquez, Eduardo Gerardo Pérez Tijerina
Instituto Tecnológico de Tijuana (ITT)
<i>Corr. author: carlos.guerrerop@uanl.edu.mx</i> |
| 14:00 | Benchmarking AI Models for Intrusion Detection and Malware Classification
Ricardo Rioda Santiago, Antonio Contreras Ortiz, Kristoffer Ancona, Miguel Lopez-Montiel
ITJuana Labs
<i>Corr. author: miguel.lopez@itj.com</i> |
| 14:20 | Solving a Hamiltonian Function of Split Transmon Qubits Capacitively Coupled with a Resonator in a Cavity Quantum Electrodynamics
Afees Olamilekan Tihamiyu, Seyi Tope Ogunji, Olugbadebo Gbolade Fredrick, Ekpenyong Ukut Ini, Asuquo Afangideh Aniekan, Moises Sanchez-Adame, Oscar Montiel Ross
Centro de Investigación y Desarrollo de Tecnología Digital (CITEDI) IPN
<i>Corr. Author: oross@ipn.mx</i> |
| 14:40 | CLOSING DAY 2 |